

Executive Business Report

Retail Sales Forecasting & Business Intelligence Dashboard

Prepared for:

Head of Supply Chain & Chief Financial Officer (CFO)

1. Executive Summary

During this project, historical retail sales data from 2015 to 2018 was analyzed to understand business performance, identify unusual sales patterns, forecast future demand, and segment products based on their sales behaviour. Multiple forecasting models were evaluated, and Prophet was selected as the primary forecasting model because it achieved the lowest forecasting errors (MAE and RMSE). Anomaly detection and product demand segmentation were also performed to support better inventory planning and business decision-making. The developed Streamlit dashboard enables managers to interactively explore forecasts, anomalies, and product demand insights in an easy-to-understand interface.

2. Key Findings from EDA and Forecasting

Exploratory Data Analysis (EDA)

- Technology generated the highest overall revenue (approximately \$830K) among all product categories, followed by Furniture and Office Supplies.
- The West region consistently achieved the highest sales and demonstrated the strongest growth throughout the analysis period.
- Sales showed a clear seasonal pattern, with demand increasing significantly during the last quarter of each year, especially in November and December.
- Overall business performance improved steadily between 2015 and 2018, indicating continuous growth despite normal month-to-month fluctuations.

Forecasting Results

Three forecasting models (SARIMA, Prophet, and XGBoost) were evaluated.

Model	MAE	RMSE
SARIMA	14,437.20	19,386.18
Prophet	5,770.42	7,272.00
XGBoost	8,903.40	13,351.83

Based on the evaluation, Prophet was selected because it produced the lowest forecasting errors and generated the most reliable future sales predictions.

3. Three-Month Sales Forecast

The selected Prophet model predicts the following sales for the next three months:

Month	Forecast	Confidence Interval
January 2019	\$42,991	\$33,384 – \$52,406
February 2019	\$31,248	\$21,642 – \$40,360
March 2019	\$81,267	\$72,260 – \$90,545

The forecast suggests a temporary decline during February followed by a strong recovery in March. The confidence intervals indicate the expected range within which future sales are likely to occur.

4. Top Three Anomalies Detected

The anomaly detection process identified several unusually high sales weeks.

Date	Sales
22 March 2015	\$37,703.67
13 September 2015	\$29,959.14
08 October 2017	\$28,412.10

These unusual sales spikes are likely associated with promotional campaigns, seasonal demand, or large customer purchases. Such anomalies should be monitored separately because they may not represent normal customer behaviour.

5. Product Demand Segmentation

Products were grouped into four demand segments using K-Means clustering.

Segment	Recommended Stocking Strategy
High-Value Premium Products	Maintain moderate inventory levels while ensuring product availability.
Low Volume, Stable Demand	Keep minimum but consistent inventory through routine replenishment.
High Volume, Stable Demand	Maintain high inventory levels with regular restocking to prevent stock-outs.
Rapidly Growing Demand	Closely monitor demand trends and gradually increase inventory while updating forecasts frequently.

This segmentation enables inventory decisions to be tailored according to each product group's sales behaviour.

6. Business Recommendations

- **Prioritize inventory for Technology products and the West region.** Both consistently generated the highest sales and contribute significantly to overall revenue.
- **Increase inventory before the holiday season.** Historical sales patterns show strong demand during October to December, particularly in November. Preparing inventory in advance can reduce stock shortages.
- **Use forecasting to optimize inventory planning.** Integrating Prophet forecasts into monthly planning can reduce overstocking while ensuring sufficient inventory during periods of increased demand.

7. Risk / Limitation

The forecasting models were trained using historical sales data from 2015–2018. Unexpected events such as economic changes, new competitors, supply chain disruptions, or major promotional campaigns are not reflected in the historical dataset and may reduce forecasting accuracy. Therefore, forecasts should be updated regularly as new sales data becomes available.